Inclusion–exclusion exercises

1 • Three divisibility conditions

How many integers in 1,…,200 are divisible by none of 2,3,5? List all orders of intersections. Explain why multiplying independent-event probabilities is unjustified.

2 • Exactly two owners recover their passes

Assign six distinct passes bijectively to their six owners. Count assignments with exactly two correct owners, using disjoint exhaustive classes.

3 • Two checks for derangements

Use the general cycle-deletion/contraction bijection to derive the derangement recurrence. Find D_5,D_6 and check D_5 by inclusion–exclusion. Explain the empty-permutation base case.

4 • A full forbidden board

For the full 3×3 board B, find the rook polynomial and the number of permutations avoiding B. Explain each coefficient's combinatorial meaning.

5 • Rectangular assignments

Assign three named jobs to four named people, at most one job per person. Forbidden cells are (1,1),(2,2),(3,3),(3,4). Find the rook polynomial and legal assignment count; list every legal assignment to check the example.

6 • Conditions for board multiplication

Prove $r_k(B_1 \cup B_2) = \sum_i r_i(B_1) r_{k-i}(B_2)$ when boards share neither rows nor columns. Does cell disjointness alone suffice? Give a smallest counterexample and identify the failed step.

7 • Reconstruct the hit distribution

A universe has 30 objects and three properties with intersection sums S_1=26,S_2=12,S_3=2. Find counts for exactly 0,1,2,3 and at least two hits. Reconstruct every S_k from your answer. Give union bounds using only S_1,S_2.

Inclusion–exclusion exercises (continued)

8 • Why truncated bounds have directions

For one object satisfying c properties, prove the truncated alternating binomial identity and deduce finite-universe Bonferroni inequalities. With possibly negative object weights, which statement survives and which fails? Give an explicit counterexample.

9 • Bounded integer solutions

Count integer solutions of $x_1+x_2+x_3=8$ with $0 \le x_i \le 3$. Compute single and double bound violations separately and check using $y_i=3-x_i$. Explain why a general bound b_i requires shifting by b_i+1.

10 • Noncoprime divisors

Count integers in 1,…,60 divisible by neither 4 nor 6. What wrong answer results from treating the intersection as multiples of 24? Derive $\phi(36)$ by inclusion–exclusion, explaining prime powers.

11 • Full ladder-board distribution

For the 4×4 board $B=\{(1,1),(1,2),(2,2),(2,3),(3,3),(3,4)\}$, derive every rook coefficient and exact-hit permutation count h_t for t=0,…,4. List all avoiding permutations and check $\sum_t t h_t$.

12 • Circular contraction and orientation

Seat eight distinct people in four specified pairs around a circle, identifying rotations and distinguishing reflections. Count arrangements where no pair is adjacent. Prove the intersection count for k adjacent pairs and explain why the same orientation factor fails for a two-person circle.

13 • No isolates versus connectivity

Count labeled simple undirected four-vertex graphs without isolated vertices, then connected graphs. Explain intersections of specified isolated-vertex events. Classify possible component sizes for disconnected no-isolate graphs.

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Inclusion–exclusion exercises (continued)

14 • Rook dynamic-programming states

Give and prove the row recurrence with occupied-column-set states, including arithmetic and space costs. Use one row with cells (1,1),(1,2) to show how in-place updates toward larger sets can create an illegal placement. Use deletion/contraction to compute the polynomial for {(1,1),(1,2),(2,1)}.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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